

WEEK 8: PRACTICAL

Activities

1. Create VLANs.

```
Switch>enable
Switch#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name LECTURERS
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name STUDENTS
Switch(config-vlan)#exit
Switch(config)#
```

2. Assign ports to VLANs.

```
Switch#config term
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastethernet 0/1-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fastethernet 0/4-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range fastethernet 0/7-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#end
```

3. Verify VLAN membership.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	ADMIN	active	Fa0/1, Fa0/2, Fa0/3
20	LECTURERS	active	Fa0/4, Fa0/5, Fa0/6
30	STUDENTS	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

4. Test communication.

PC1 = PC2 ; PC1 = PC3 (ADMINISTRATION) – VLAN 10

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=3ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

PC4=PC5 ; PC4=PC6 (LECTURERS) -VLAN 20

```

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

PC7=PC8 ; PC7=PC9 ; PC7=PC10 (STUDENTS) -VLAN 30

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.12

Pinging 192.168.30.12 with 32 bytes of data:

Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.13

Pinging 192.168.30.13 with 32 bytes of data:

Reply from 192.168.30.13: bytes=32 time<1ms TTL=128
Reply from 192.168.30.13: bytes=32 time<1ms TTL=128
Reply from 192.168.30.13: bytes=32 time<1ms TTL=128
Reply from 192.168.30.13: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.30.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

```

5. Document VLAN design.

VLAN	DEPARTMENT	ASSIGNED PCs	SWITCH PORTS	NETWORK
10	ADMINISTRATION	PC1-PC3	Fa0/1-Fa0/3	192.168.10.0/24
20	LECTURERS	PC4-PC6	Fa0/4-Fa0/6	192.168.20.0/24
30	STUDENTS	PC7-PC8	Fa0/7-Fa0/10	132.168.30.0/24

Verification

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	ADMIN	active	
20	LECTURERS	active	
30	STUDENTS	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Copy

Paste

TOPOLOGY

